SIMATS ENGINEERING

Engineer to Excel

SIMATS

Saveetha Institute of Medical And Technical Sciences

(Declared as Deemed to be University under Section 3 of UGC Act 1956)

DESIGN AND ANALYSIS OF A SMART LOGISTICS NETWORK OPTIMIZATION SYSTEM

CSA0620-Design And Analysis Of Algorithms

Guided by:

DR. Parimala Kanaga Devan K
Professor CSE

Presented by :

P. Sai Ganesh(192525111)
S. Chandra Siddhardha(192525222)

INTRODUCTION

- Logistics involves transporting goods efficiently between multiple locations.
- Poor route selection can increase delivery time, distance, and transportation cost.
- The logistics network can be represented as a weighted graph.
- Graph algorithms can identify efficient routes between warehouses and customers.
- The proposed system provides intelligent route optimization and delivery planning.

PROBLEM STATEMENT

- Logistics companies waste significant time and fuel due to inefficient routing
- Manual route planning causes delivery delays and uneven vehicle utilization
- Existing systems rarely consider shipment priority, distance, and vehicle capacity together
- Without route optimization, drivers cannot reach destinations efficiently
- Need for an intelligent system that processes shipments and optimizes routes automatically

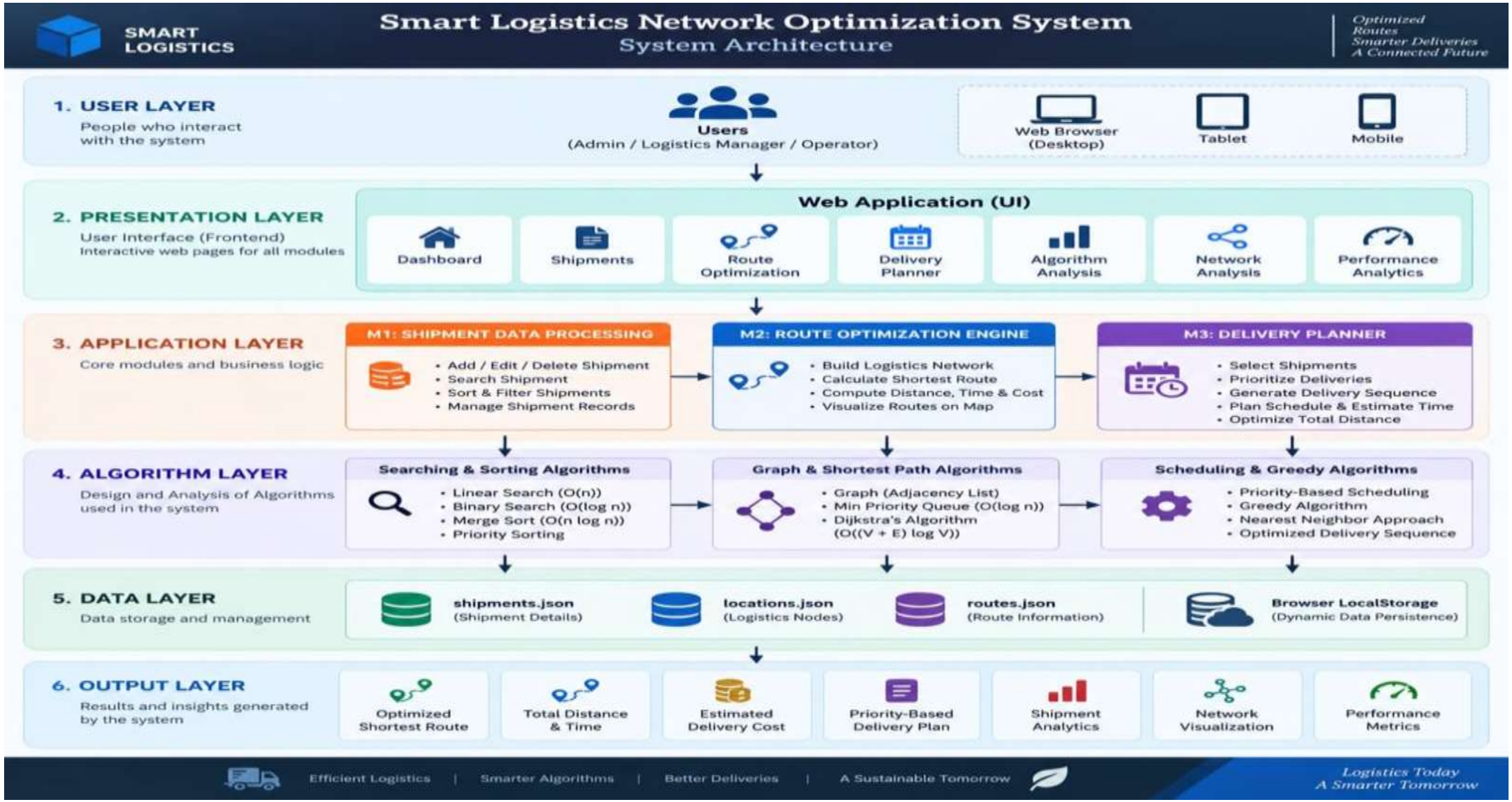
OBJECTIVES

- Design a system to manage shipment data and vehicle availability
- Process incoming shipments and allocate the optimal vehicle automatically
- Optimize delivery routes using the shortest path with Dijkstra's Algorithm
- Analyze system performance using delivery time and utilization metrics
- Evaluate the advantages and limitations of each technique used

LITERATURE REVIEW

Author(s) & Year	Title	Algorithms Discussed	Methodology	Key Findings
Khan et al. (2024)	Vehicle Routing Problems in Multi-Echelon Distribution Networks	VRP, Heuristics, Metaheuristics	Systematic review of multi-echelon routing models	Identified routing, scheduling and sustainable logistics as key areas. (sciencedirect.com)
Adamo et al. (2024)	Recent Advances in Time-Dependent Vehicle Routing	Time-Dependent VRP, Dynamic Routing	Reviewed routing under changing travel-time conditions	Emphasized dynamic route optimization for modern logistics.
Monemi et al. (2025)	Graph Convolutional Networks for Logistics Optimization	GCN, RL, Graph Optimization	Survey of graph-based routing and scheduling methods	GCNs improve scalability for complex logistics problems. (sciencedirect.com)
Kuliavets & Burov (2025)	Delivery Delay Prediction and Route Optimization Using GNNs	GNN, Dynamic Routing	Combined GNN delay prediction with real-time route optimization	Improved prediction of delays and adaptive route planning. (science.lpnu.ua)
Zabraoui et al. (2026)	AI-Driven Optimization of Vehicle Routing Problems in Supply Chain	GNN, RL, VRP	Integrated GNNs and reinforcement learning for VRP	AI-based routing improves adaptability, efficiency and scalability. (doi.org)

SYSTEM ARCHITECTURE



TECHNOLOGIES USED

Category	Technology / Tools
Programming Language	Python
Core Algorithms	Dijkstra’s Algorithm, Greedy Algorithm
Data Structures	Priority Queue, Hashing, Graphs (BFS/DFS)
Frontend	HTML, CSS, JavaScript
Platform	Windows

MODULE 1:SHIPMENT DATA PROCESSING

- Maintain shipment records with source, destination, and priority
- Track real-time shipment status: pending, in-transit, or delivered
- Store shipment weight, destination distance, and vehicle type compatibility
- Use hashing for constant-time shipment lookup by shipment ID
- Update shipment status instantly on dispatch and delivery
- Support multiple depots with independent shipment queues

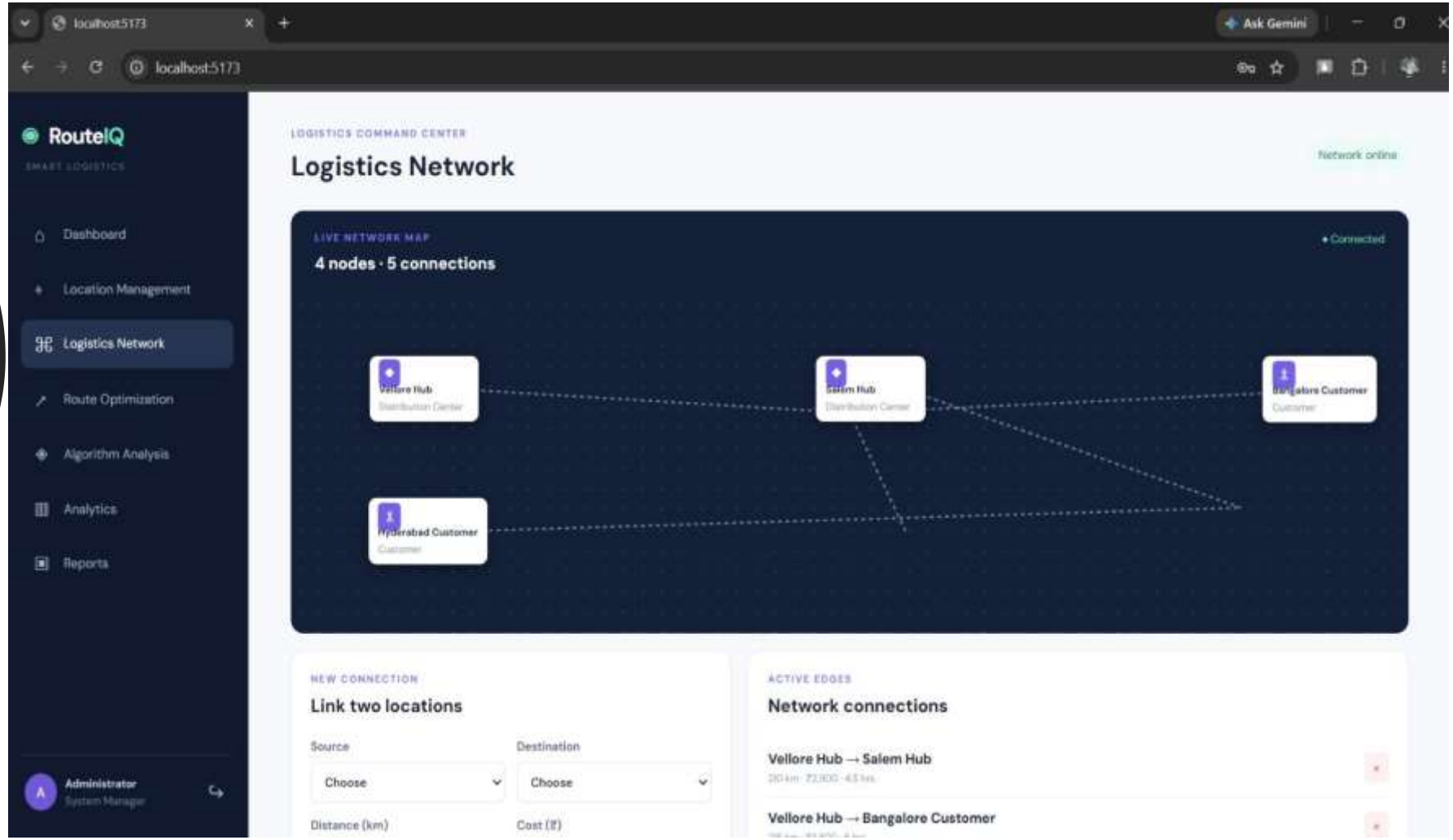
MODULE 2: ROUTE OPTIMIZATION ENGINE

- Register vehicle number, capacity, and current location at the depot
- Compute the optimal delivery route using a Greedy Algorithm
- Rank candidate routes with a Priority Queue by distance and shipment priority
- Reserve the route instantly to prevent conflicting assignments
- Log route records with distance and timestamp for analytics

MODULE-3: DELIVERY PLANNER

- **Order Management:** Collects customer orders, delivery locations, quantities, and time requirements.
- **Vehicle Assignment:** Assigns suitable vehicles based on capacity, availability, and delivery requirements.
- **Route Optimization:** Selects the shortest and most efficient routes to reduce travel time and fuel usage.
- **Delivery Scheduling:** Creates optimized delivery schedules and determines the best sequence of stops.
- **Real-Time Monitoring:** Tracks delivery progress and provides alerts for delays, route changes, and other issues.

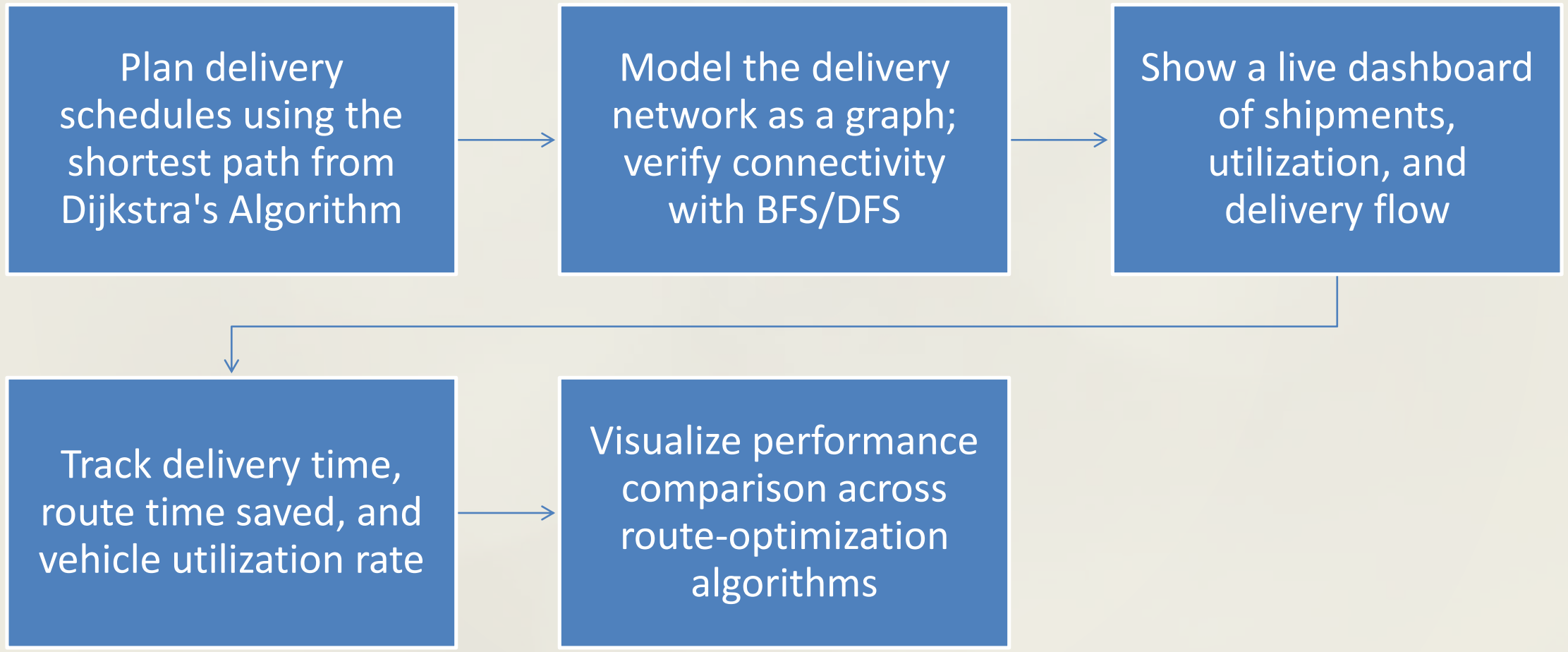
OUTPUT


localhost:5173
Ask Gemini
RouteIQ
Dashboard
Location Management
Logistics Network
Route Optimization
Algorithm Analysis
Analytics
Reports
Administrator
LOGISTICS COMMAND CENTER
Logistics Network
network online
LIVE NETWORK MAP
4 nodes · 5 connections
• Connected
Vellore Hub
Salem Hub
Hyderabad Customer
NEW CONNECTION
Link two locations
Source
Destination
Choose
Choose
Distance (km)
Cost (₹)
ACTIVE EDGES
Network connections
Vellore Hub → Salem Hub
Vellore Hub → Bangalore Customer

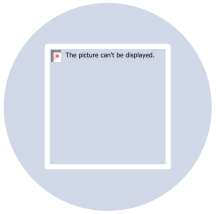
Advantages

- Reduces transportation and fuel costs
- Minimizes delivery time and delays
- Improves route efficiency
- Optimizes resource utilization
- Supports priority-based deliveries
- Improves customer satisfaction

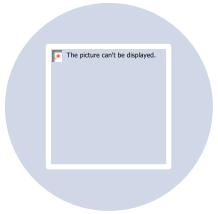
DELIVERY PLANNER & DASHBOARD



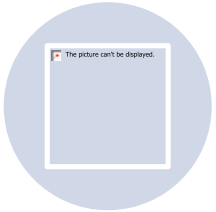
PROPOSED SYSTEM



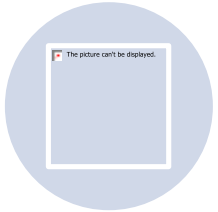
Uses **graph-based algorithms** to model the logistics network.



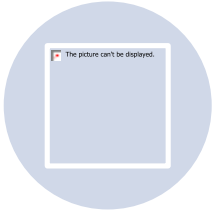
Finds **shortest and optimal routes** between locations.



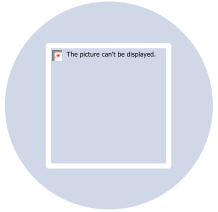
Considers **distance, cost, time, and traffic conditions**.



Uses **priority queues** for urgent and priority deliveries.



Optimizes **vehicle and warehouse utilization**.



Provides **smart and efficient delivery planning**.

CONCLUSION

- The proposed system applies graph algorithms to logistics optimization.
- It identifies efficient routes between warehouses, hubs, and customers.
- Dijkstra, A*, and Floyd–Warshall provide different approaches to route optimization.
- Priority Queue and Greedy techniques improve delivery planning.
- Algorithm analysis helps select an efficient solution based on network requirements.
- The system can support faster, cost-effective, and intelligent logistics planning.

REFERENCES

- **Khan et al. (2024)**, *A Systematic Review of Vehicle Routing Problems and Models in Multi-Echelon Distribution Networks*, **Supply Chain Analytics, 7, 100072**.
- **Adamo, T., Gendreau, M., Ghiani, G., & Guerriero, E. (2024)**, *A Review of Recent Advances in Time-Dependent Vehicle Routing*, **European Journal of Operational Research, 319(1)**.
- **Nourmohammadi, Z., Hu, B., Rey, D., & Saberi, M. (2025)**, *A Data-Driven Preference Learning Approach for Multi-Objective Vehicle Routing Problems in Last-Mile Delivery*, **Transportation Research Part C, 174, 105101**.
- **Monemi, R. N., et al. (2025)**, *Graph Convolutional Networks for Logistics Optimization: A Survey of Scheduling and Operational Applications*, **Transportation Research Part E, 197, 104083**.
- **Zhou, F., Lischka, A., Kulcsár, B., Wu, J., Chehreghani, M. H., & Laporte, G. (2025)**, *Learning for Routing: A Guided Review of Recent Developments and Future Directions*, **Transportation Research Part E, 202, 104278**.





Thank you!